TOPS TECH Task-8 (CS)

Task-8 Check Live system, open ports, Services

1. **Use the netstat command on your Windows or Linux system to list all currently open ports and their associated services, then take a screenshot of the output.**

   Ans:

   List Currently Open Ports and Associated Services

   Objective:
   - To identify the currently open ports and their associated services running on the system using the Netstat command.

   Command:
   → netstat -ano

   ➢ The netstat -ano command displays all active network connections, listening ports, protocol information, connection states, and the Process ID (PID) associated with each connection. It is commonly used for network monitoring, troubleshooting, and security analysis.

   Observations:

   - The command output showed multiple active and listening ports on the system.
   - These ports are used by various Windows services and applications for network communication.

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The following open ports were identified:

Port Number	**Protocol**	**State**	**Description**
135	TCP	LISTENING	Microsoft RPC (Remote Procedure Call) Service
445	TCP	LISTENING	SMB (Server Message Block) File Sharing Service
5040	TCP	LISTENING	Windows Push Notification Service
7680	TCP	LISTENING	Windows Update Delivery Optimization
139	TCP	LISTENING	NetBIOS Session Service
2020	TCP	LISTENING	Local Application Service
2021	TCP	LISTENING	Local Application Service
2022	TCP	LISTENING	Local Application Service

- ⇨ Several connections were found in the ESTABLISHED state, indicating active communication between applications and remote systems.
- ⇨ Some connections were in TIME_WAIT and CLOSE_WAIT states, showing recently closed or terminating network sessions.
- ⇨ Multiple high-numbered ports (49664–49669) were observed, which are dynamically allocated by Windows for internal communication.

TOPS TECH Task-8 (CS)

→ Using the netstat -ano command, the currently active ports, listening services, and associated process IDs were successfully identified.

→ The system was found to be running several important Windows networking services, including RPC, SMB, NetBIOS, and Windows Update services.

→ This information is useful for system administration, network troubleshooting, and security assessment, as it helps identify which services are accepting network connections on the system.

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8655]
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C:\Users\bharg>netstat -ano

Active Connections

  Proto  Local Address          Foreign Address        State           PID
  TCP    0.0.0.0:135            0.0.0.0:0              LISTENING       1312
  TCP    0.0.0.0:445            0.0.0.0:0              LISTENING       4
  TCP    0.0.0.0:5040           0.0.0.0:0              LISTENING       5428
  TCP    0.0.0.0:7680           0.0.0.0:0              LISTENING       6916
  TCP    0.0.0.0:22321          0.0.0.0:0              LISTENING       8
  TCP    0.0.0.0:49664          0.0.0.0:0              LISTENING       972
  TCP    0.0.0.0:49665          0.0.0.0:0              LISTENING       960
  TCP    0.0.0.0:49666          0.0.0.0:0              LISTENING       1880
  TCP    0.0.0.0:49667          0.0.0.0:0              LISTENING       2852
  TCP    0.0.0.0:49668          0.0.0.0:0              LISTENING       5652
  TCP    0.0.0.0:49669          0.0.0.0:0              LISTENING       5232
  TCP    127.0.0.1:2020         0.0.0.0:0              LISTENING       12092
  TCP    127.0.0.1:2020         127.0.0.1:59473        ESTABLISHED     12092
  TCP    127.0.0.1:2021         0.0.0.0:0              LISTENING       6188
  TCP    127.0.0.1:2021         127.0.0.1:22328        ESTABLISHED     6188
  TCP    127.0.0.1:2022         0.0.0.0:0              LISTENING       6208
  TCP    127.0.0.1:2022         127.0.0.1:22330        ESTABLISHED     6208
  TCP    127.0.0.1:12965        127.0.0.1:12966        ESTABLISHED     14696
  TCP    127.0.0.1:12966        127.0.0.1:12965        ESTABLISHED     14696
  TCP    192.168.31.193:139     0.0.0.0:0              LISTENING       4
  TCP    192.168.31.193:12957   20.190.146.39:443      TIME_WAIT       0
  TCP    192.168.31.193:12959   148.113.35.154:443     TIME_WAIT       0
  TCP    192.168.31.193:12964   34.221.249.61:443      CLOSE_WAIT      10960
  TCP    [::]:135               [::]:0                 LISTENING       1312
  TCP    [::]:445               [::]:0                 LISTENING       4
  TCP    [::]:7680              [::]:0                 LISTENING       6916
  TCP    [::]:22321             [::]:0                 LISTENING       8
  TCP    [::]:49664             [::]:0                 LISTENING       972
  TCP    [::]:49665             [::]:0                 LISTENING       960
  TCP    [::]:49666             [::]:0                 LISTENING       1880
  TCP    [::]:49667             [::]:0                 LISTENING       2852
  TCP    [::]:49668             [::]:0                 LISTENING       5652
  TCP    [::]:49669             [::]:0                 LISTENING       5232
  TCP    [::1]:49670            [::]:0                 LISTENING       6496
  TCP    [2409:4090:1056:ee01:b9b3:42af:b025:9589]:12961  [2603:1040:5:3::a]:443                 ESTABLISHED   13668
  TCP    [2409:4090:1056:ee01:b9b3:42af:b025:9589]:27425  [2a03:2880:f379:120:face:b00c:0:167]:522ABLISHED   15116
  TCP    [2409:4090:1056:ee01:b9b3:42af:b025:9589]:49410  [2603:1040:a06:6::]:443                ESTABLISHED   7064
  TCP    [2409:4090:1056:ee01:b9b3:42af:b025:9589]:49411  [2603:1040:a06:6::]:443                ESTABLISHED   7064
  TCP    [2409:4090:1056:ee01:b9b3:42af:b025:9589]:58401  [2404:6800:4000:1025::bc]:5228          ESTABLISHED   12964
  UDP    0.0.0.0:123            *:*
  UDP    0.0.0.0:5050           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5355           *:*
```

2. **Check which applications are using ports 80 and 443 on your system by using the command line tool (netstat, lsof, or Resource Monitor), and write down the process names you find.**

Ans:

Identify Applications Using Ports 80 and 443

Objective:

- To determine which applications are using ports 80 (HTTP) and 443 (HTTPS) on the system.

Commands:

→ netstat -ano | findstr :80
→ netstat -ano | findstr :443
→ tasklist | findstr chrome

- ➢ The netstat command was used to identify active network connections on ports 80 and 443.
- ➢ The associated application was verified using the tasklist command.

Observations:

Port 80 (HTTP):

- Network activity related to web communication was detected during the inspection.

Port 443 (HTTPS):

- Port 443 was being used by web browser and secure network services for encrypted communication.

Results:

Port Number	Service Type	Process Name	Status
80	HTTP	Not in Use	Closed / No Active Process
443	HTTPS	Chrome.exe (or Browser Service)	Active

Purpose of Ports:

Port 80:

- Used for standard HTTP web traffic and communication between web clients and servers.

Port 443:

- Used for HTTPS secure web traffic and encrypted communication over the internet.

  - ⇨ The system was checked for applications using ports 80 and 443.
  - ⇨ Active network communication was observed, and Chrome.exe was identified as one of the applications using secure HTTPS connections through port 443.
  - ⇨ These ports are essential for normal web browsing and internet-based services.

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Commands:

→ netstat -ano | findstr :80

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>netstat -ano | findstr :80
  TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:17601  [2404:6800:4009:80a::200e]:443  ESTABLISHED     17396
  TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:26323  [2620:1ec:33::11]:80    TIME_WAIT       0
  TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:48064  [2603:1046:c04:80c::2]:443  TIME_WAIT       0

C:\Windows\System32>
```

Commands:

→ netstat -ano | findstr :443

```
Administrator: Command Prompt

C:\Windows\System32>netstat -ano | findstr :443
  TCP    192.168.31.165:40447   3.110.26.57:443        CLOSE_WAIT      17396
  TCP    192.168.31.165:63403   140.82.113.26:443      ESTABLISHED     17396
  TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:2064  [2603:1046:c04:104b::2]:443  ESTABLISHED     10988
  TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:9360  [2603:1046:900:54::2]:443  ESTABLISHED     10988
  TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:17601  [2404:6800:4009:80a::200e]:443  ESTABLISHED     17396
  TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:18087  [2600:140f:1c00::b854:e8e9]:443  ESTABLISHED     19444
  TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:22532  [2404:6800:4000:1025::54]:443  ESTABLISHED     17396
  TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:26324  [2603:1063:1f:103::365:7ea3]:443  ESTABLISHED     5124
  TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:32824  [2603:1046:c04:1048::2]:443  ESTABLISHED     10264
  TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:39691  [2606:4700:8d71:d8b7:9a59:649:ccf5:ac50]:443  ESTABLISHED     17396
  TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:41232  [2404:6800:4000:1025::54]:443  ESTABLISHED     17396
  TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:44395  [2606:4700:8d71:d8b7:9a59:649:ccf5:ac50]:443  ESTABLISHED     17396
  TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:49408  [2603:1040:a06:6::]:443  ESTABLISHED     5956
  TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:51199  [2603:1063:2001:2008::365:ff1]:443  ESTABLISHED     10988
  TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:51200  [2603:1063:2001:2008::365:ff1]:443  ESTABLISHED     10988
  TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:51201  [2603:1063:2001:1943::365:ff1]:443  ESTABLISHED     10988
  TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:52563  [2606:4700:8ca1:d8b7:9aa7:44:ccf5:ac50]:443  ESTABLISHED     17396
  TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:54851  [2600:1901:0:47fc::]:443  ESTABLISHED     17396
  TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:55003  [2404:6800:4000:1025::54]:443  ESTABLISHED     17396
  TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:55531  [2404:6800:4002:831::200a]:443  ESTABLISHED     17396
  TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:56329  [2603:1063:2001:2004::365:ff1]:443  ESTABLISHED     12324
  TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:56332  [2603:1063:2001:2008::365:ff1]:443  ESTABLISHED     12324
  TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:61775  [2606:4700:8d72:7cbc:c243:643:5ff2:75c4]:443  ESTABLISHED     17396
  TCP    [2409:4090:1056:ee01:9479:6684:4bea:606f]:63314  [2603:1046:c04:1401::2]:443  ESTABLISHED     10264

C:\Windows\System32>_
```

Commands:

→ tasklist | findstr chrome

```
Administrator: Command Prompt

C:\Windows\System32>tasklist | findstr chrome
chrome.exe                   16892 Console                    1   1,50,412 K
chrome.exe                   17096 Console                    1      2,396 K
chrome.exe                   17376 Console                    1   1,17,448 K
chrome.exe                   17396 Console                    1     33,792 K
chrome.exe                   16388 Console                    1      7,284 K
chrome.exe                   14216 Console                    1     58,580 K
chrome.exe                   14424 Console                    1     26,524 K
chrome.exe                   15868 Console                    1     23,816 K
chrome.exe                   14000 Console                    1   2,69,304 K
chrome.exe                   14212 Console                    1     45,748 K
chrome.exe                    6460 Console                    1     94,352 K
chrome.exe                   17028 Console                    1     20,400 K
chrome.exe                   13140 Console                    1     38,132 K
chrome.exe                   23468 Console                    1     49,164 K

C:\Windows\System32>
```

3. **Scan your own system for open ports using Nmap or Zenmap and identify at least three services running on those ports.

Hint: If you don't have Nmap installed, download it from nmap.org or use Zenmap GUI for easier results.**

Ans:

Scan the System for Open Ports Using Nmap

Objective:

- To scan the local system for open ports and identify the services running on those ports using Nmap.

Command:

→ nmap localhost

- Nmap (Network Mapper) was used to scan the local machine and identify open ports along with the services associated with them.
- Nmap is a widely used network scanning and security auditing tool that helps detect active hosts, open ports, and running services.

Observations:

- The scan was performed successfully on the localhost (127.0.0.1). The system was found to be active and responding. Nmap identified two open TCP ports and their associated services.

Results:

Port Number	**State**	**Service**
135/tcp	Open	MSRPC
445/tcp	Open	Microsoft-DS

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Service Details:

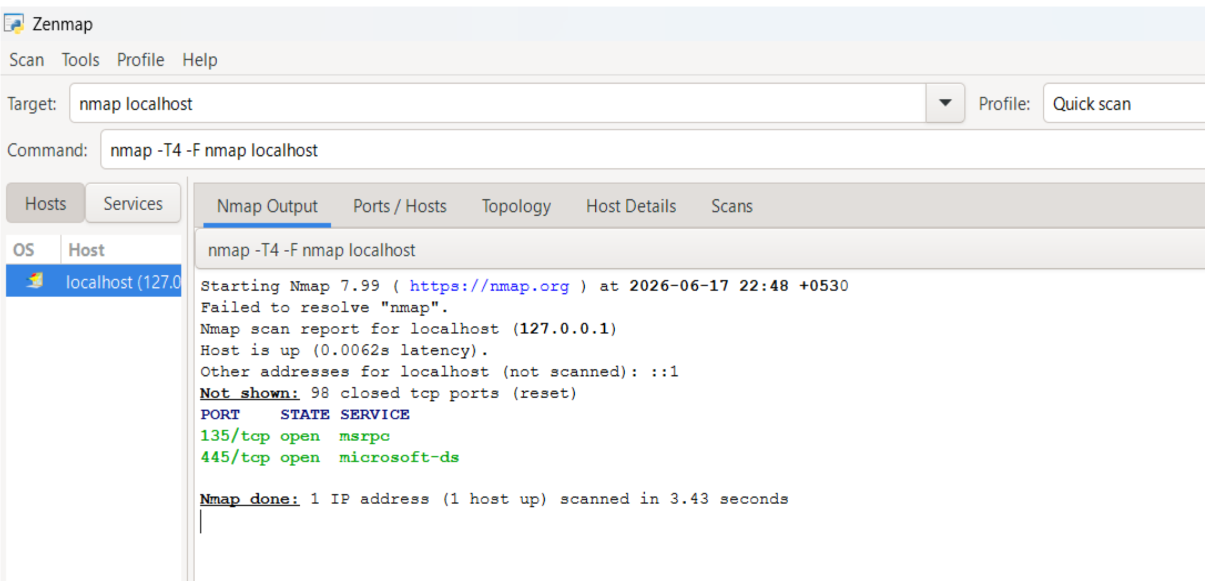
MSRPC (Port 135):

- Microsoft Remote Procedure Call (MSRPC) is used by Windows operating systems to allow communication between software applications and system services across a network.

Microsoft-DS (Port 445):

- Microsoft Directory Services (Microsoft-DS) is used for file sharing, printer sharing, and other network communication services through the SMB (Server Message Block) protocol.

  - ⇨ The local system was successfully scanned using Nmap. Two open ports, 135 and 445, were identified along with their associated services.
  - ⇨ These services are commonly used by Windows systems for internal communication and file-sharing operations. The scan provided useful information about the active network services running on the system.



4. Pick any app you use daily (like Spotify, WhatsApp Desktop, or a local web server) and check which ports it opens on your system while running. List the port numbers and explain in one line what each port is typically used for.

Ans:

Check Ports Used by a Running Application

Objective:

- To identify the ports used by a running application on the system and understand the purpose of those ports.

Selected Application:

→ Google Chrome

Commands:

→ netstat -ano

→ tasklist | findstr chrome

- Google Chrome was selected as the application for this task.
- The netstat command was used to identify active network connections, while the tasklist command was used to verify the running Chrome processes.

Observations:

- Multiple instances of Chrome.exe were found running on the system.
- Active network connections associated with Chrome were observed, particularly on port 443, which is commonly used for secure web communication.

Results:

Application	Port Number	Protocol	Purpose
Google Chrome	443	HTTPS	Secure and encrypted web communication
Google Chrome	Dynamic Client Ports	TCP	Establishing connections to remote web servers

Port Details:

Port 443 (HTTPS):

- This port is used for secure communication between the web browser and websites.
- Data transferred through this port is encrypted using SSL/TLS protocols, ensuring privacy and security.

Dynamic Client Ports:

- Chrome also uses temporary client-side ports assigned by the operating system to establish outbound connections with remote servers.

⇨ Google Chrome was analyzed as a running application on the system.
⇨ The application was actively using port 443 for secure web browsing and encrypted communication.
⇨ Dynamic client ports were also observed for establishing network sessions with web servers. This demonstrates how modern web browsers rely on multiple network connections for secure internet access.

Command:

→ netstat -ano

```
C:\Windows\System32>netstat -ano

Active Connections

  Proto  Local Address          Foreign Address        State           PID
  TCP    0.0.0.0:135            0.0.0.0:0              LISTENING       1668
  TCP    0.0.0.0:445            0.0.0.0:0              LISTENING       4
  TCP    0.0.0.0:1521           0.0.0.0:0              LISTENING       5448
  TCP    0.0.0.0:1947           0.0.0.0:0              LISTENING       4848
  TCP    0.0.0.0:5040           0.0.0.0:0              LISTENING       9704
  TCP    0.0.0.0:7680           0.0.0.0:0              LISTENING       16256
  TCP    0.0.0.0:8885           0.0.0.0:0              LISTENING       9076
  TCP    0.0.0.0:8886           0.0.0.0:0              LISTENING       9076
  TCP    0.0.0.0:49664          0.0.0.0:0              LISTENING       1368
  TCP    0.0.0.0:49665          0.0.0.0:0              LISTENING       1204
  TCP    0.0.0.0:49666          0.0.0.0:0              LISTENING       2092
  TCP    0.0.0.0:49667          0.0.0.0:0              LISTENING       3188
  TCP    0.0.0.0:49668          0.0.0.0:0              LISTENING       4176
  TCP    0.0.0.0:49684          0.0.0.0:0              LISTENING       1316
  TCP    127.0.0.1:5354         0.0.0.0:0              LISTENING       4484
  TCP    127.0.0.1:7080         127.0.0.1:7081         ESTABLISHED     3828
  TCP    127.0.0.1:7081         127.0.0.1:7080         ESTABLISHED     3828
  TCP    127.0.0.1:8884         0.0.0.0:0              LISTENING       4
  TCP    127.0.0.1:22350        0.0.0.0:0              LISTENING       4552
  TCP    127.0.0.1:22350        127.0.0.1:22665        ESTABLISHED     4552
  TCP    127.0.0.1:22352        0.0.0.0:0              LISTENING       4536
  TCP    127.0.0.1:22665        127.0.0.1:22350        ESTABLISHED     12936
  TCP    127.0.0.1:49678        0.0.0.0:0              LISTENING       5448
  TCP    192.168.31.165:139     0.0.0.0:0              LISTENING       4
  TCP    192.168.31.165:2739    20.184.175.23:443      ESTABLISHED     3424
  TCP    192.168.31.165:8974    192.168.31.1:53        TIME_WAIT       0
  TCP    192.168.31.165:13595   3.110.26.57:443        ESTABLISHED     17396
```

Command:

→ tasklist | findstr chrome

```
C:\Windows\System32>tasklist | findstr chrome
chrome.exe                   16892 Console                    1   2,02,532 K
chrome.exe                   17096 Console                    1      2,044 K
chrome.exe                   17376 Console                    1   2,44,620 K
chrome.exe                   17396 Console                    1     38,344 K
chrome.exe                   16388 Console                    1      9,128 K
chrome.exe                   14424 Console                    1     96,536 K
chrome.exe                   15868 Console                    1     92,516 K
chrome.exe                   14000 Console                    1   3,24,464 K
chrome.exe                   14212 Console                    1     72,272 K
chrome.exe                    6460 Console                    1     31,320 K
chrome.exe                   17028 Console                    1     15,368 K
chrome.exe                   13140 Console                    1     15,668 K
chrome.exe                   10000 Console                    1   2,11,952 K
chrome.exe                    7040 Console                    1   4,16,580 K
chrome.exe                   16328 Console                    1     62,632 K

C:\Windows\System32>
```